

Report
on
“Comparative analysis of fertilizers of LCB and other existing products in mungbean”
(Product: ‘Navyakosh’)



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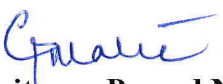


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CERTIFICATE

This is to certify that the contract research project entitled **“Comparative analysis of fertilizers of LCB and other existing products in mungbean (product ‘Navyakosh’)”** was undertaken by Division of Crop Production, ICAR– Indian Institute of Pulses Research, Kanpur as per the proposed technical programme. The report embodies the results of *Kharif* season of 2023.

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Contents

Items	Page no.
Chapter I: Introduction	3-4
Chapter II: Materials and methods	5-13
Chapter III: Results	14-20
Chapter IV: Discussion	21
Chapter V: Inferences	22
References	23-25

Pulses are an important component of Indian agricultural economy next to the cereals and oilseeds in terms of acreage, production and economic value (Choudhary, 2009). Pulses are an integral part of vegetarian diet of a large population in India. Besides being a rich source of proteins and essential amino acids; they also maintain soil fertility through biological nitrogen fixation in symbiotic association with *Rhizobium* bacteria present in their root nodules (Kumar et al., 2019). Pulses are rich sources of protein and energy but in India, these are largely cultivated under energy starved conditions, mostly on marginal and sub-marginal land and more than three-fourth of the area under pulses is still *rainfed* resulting in poor crop productivity (Choudhary, 2013). By observing the significance of pulses and to promote the pulses production across the world United Nations declared the year 2016 as “International year of Pulses”.

Recent evidences of some second-generation problems in agriculture like stagnating productivity and sustainability of the rice-wheat system, declining in total factor productivity, nutrient imbalance, and changing climate (Das et al., 2020) are a gravest threat to Indian Agriculture. There is need for shifting cropping systems and/or production practices in accordance to the resource availability, particularly with soil characterization and water availability. Inclusion of pulses in double cereal-based cropping system or replacement of some cereal crop with pulses particularly under conservation agricultural practices can be future drivers of agricultural change in the north-western Indo-Gangetic Plains of India (Nath et al., 2019).

Mungbean (*Vigna radiata* L.) is an important grain legume crop in the world comprising an area of 7.3 M ha and 5.3 Mt of production in 2017-18. The growing demand for mungbean has led to an increased global cultivated area and export (Bindumadhava et al., 2017). In India, mungbean is grown in a 4.07 M ha area with a production of 2.02 Mt (DAC, 2018). Mungbean is one of the most important legume crops in Asia (Kumar et al., 2016). It has the potential for food and nutritional security for the escalating population because of its higher tolerance to heat and drought stress. Mungbean is a candidate crop for sustainable cropping intensification and diversification in a cereal-cereal system in south Asia (Hazra et al., 2019; Nath et al., 2019). However, the global productivity of mungbean is very low with an average yield of 0.73 t ha⁻¹ (Nair et al., 2014). Reasons behind the poor productivity of mungbean are growing in less fertile soil and poor management practices. Hence, there is a scope for enhancing the yield with suitable agro-techniques and crop management practices.

Achieving high biomass at early vegetative stage is a strategy to enhance crop productivity in pulses including mungbean. The strategies for enhancing early vegetative stage include breeding approach, physiological mechanisms and agronomic management. The agronomic interventions such as application of biofertilizer (growth stimulant, anti-transpirant, growth promoter etc.), and foliar nutrition are recommended to increase yield of mungbean under limited resources and drought condition. With the availability of biofertilizers a new window has been opened for enhancing the crop yields under stress conditions. Biofertilizers have a pivotal role in crop growth, development and source to sink relationship to enhance the crop yield. However, researches are limited on impact of biofertilizers on growth, yield and performance of mungbean. Various biofertilizers are being developed for cereals, pulses and other crops. However, their efficacy varies with crop, ecology and management. Therefore, crop- and- region specific evaluation is needed to have a better performance of bio-fertilizers. Further, their efficacy varies with dose, time of application and crop morphology. Therefore, it is essential to optimize the dose and time for specific crop across ecology for higher yield and economics.

The present experiment, **“Comparative analysis of fertilizers of LCB and other existing products in mungbean (product ‘Navyakosh’)”** was undertaken with the following objectives:

- ✿ To evaluate the effect of Navyakosh fertilizer on mungbean growth, yield and profitability
- ✿ To determine the effect of Navyakosh fertilizer on soil properties

The field experiment entitled “**Comparative analysis of fertilizers of LCB and other existing products in mungbean**” was conducted during *Kharif* season (*Last week of July to September*) of 2023 to evaluate the efficacy of ‘*Navyakosh*’. The detailed account of the materials used and techniques followed during experimentation is given in this chapter.

2.1. Experimental site description and meteorological data

The experiment was conducted at D2 field of New Research Campus of ICAR- Indian Institute of Pulses Research (ICAR-IIPR), Kanpur (Uttar Pradesh). The experimental site (26°27' N, 80°14' E) falls in the subtropical humid climate with mean annual precipitation of 794 mm. **The mean maximum and minimum air temperature were 35.3 and 26.5°C, respectively during August to September 2023. Experimental site belongs to soil order *Inceptisol (Typic Ustochrept)* with sandy loam texture.** The soil was having initial pH of 8.2 (a soil-to-water ratio of 1:2). The soil is low in available nitrogen (102 kg ha⁻¹), low in available P (P₂O₅: 12.1 kg ha⁻¹), medium in K (K₂O: 195.1 kg ha⁻¹) and low in soil organic carbon (0.3%).

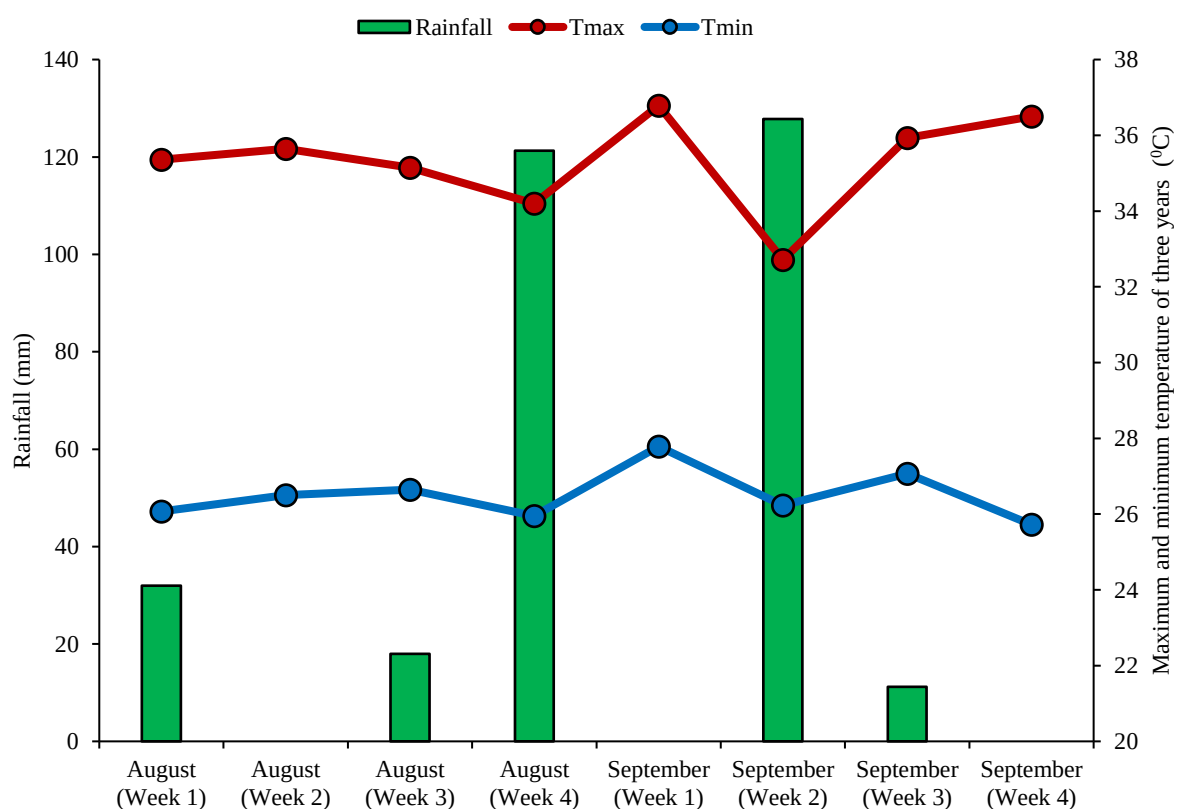


Fig. 2.1. Meteorological data of crop season of 2023

2.2. Experimental design and layout plan of the experiment

The five treatments were accommodated in randomized block design with four replications. The details of technical programme have been given below:

Treatment description

1. Recommended chemical fertilizer (20:60:40 kg/ha NPK through DAP, Urea, and MOP)
2. Navyakosh 375 kg/ha at the time land preparation
3. Farmyard manure 5 t/ha
4. Control
5. Integrated nutrient management (50% chemical fertilizers + farmyard manure 5 t/ha)

In the previous season, field was under chickpea cultivation. One ploughing was performed to improve aeration and tilth, followed by one planking to ensure proper field levelling before layout and sowing. The field experiments comprised of five treatments. Before sowing, the layout was performed for randomized block design. The manual sowing with hand plough with the same seed rate and spacing in each plot. The layout design of the experiment is given below:

Design and layout plan of the experiment:

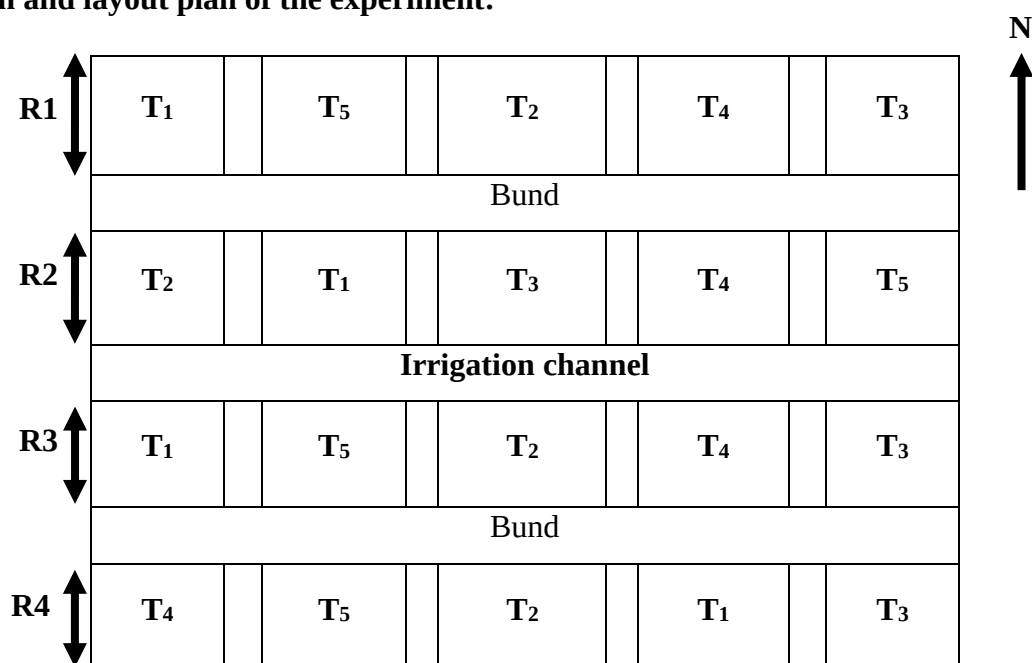


Fig. 2.2. Layout plan of the experiment

2.3. Field operations

2.3.1. Sowing

In the present experiment, mungbean variety “Virat” was sown uniformly during kharif season under conventional till condition. The date of sowing of mungbean was 21.07.2023 with randomized block design with four replications having a plot size of 8.5 m x 5 m. The mungbean crop was sown with a row to row spacing of 30 cm. After the emergence of weeds in the experimental plots, hand weeding was performed at 25 days after sowing (DAS).

2.3.2. Fertilizer application

Fertilizer with recommended dose of mungbean 20:60:40 (N:P₂O₅:K₂O kg/ha) was applied as basal as per stipulated treatment. In the farmyard manure (FYM) treated plots, FYM at 5 t/ha was applied in designated treatment as per the plot area 7 days before sowing and it was thoroughly mixed with soil. In Navyakosh treated plots, it was applied at 375 kg/ha at the time land preparation with required amount per plot. In integrated nutrient management (INM) plots, 50% NPK (10:30:20 kg/ha) and FYM 5 t/ha was applied as per requirements.

2.3.3. Irrigation

One pre-sowing irrigation was given before sowing. The number of irrigations applied in mungbean was 2 including pre-sowing irrigation.

2.3.4. Application of ‘Navyakosh’

The ‘Navyakosh’ was applied as soil application with the prescribed dose as per treatment in mungbean. It was applied at the time final land preparation and thoroughly mixed with soil. Thereafter, crop was sown.

Table 2.1. Details of cultural practices followed in mungbean.

<i>Management practices</i>	<i>Date/stage</i>
Plot size	42.5 m ² (8.5 m x 5 m)
Pre-sowing irrigation	17.07.2023
Tillage	Conventional tillage
Variety	‘Virat’
Sowing	21.07.2023

Spacing	30 (R-R) x 10 cm (P-P)
Seed rate	20 kg/ha
Weeding	16.08.2023 (at 25 days after sowing: DAS)
Observations on growth parameter	11.09.2023
Harvesting	25.09.2023



Fig. 2.3. Experimental overview

2.4. Observations on crops

a) Root length: Length was measured at 30 DAS. Five plants were randomly selected at the third rows plot from each side of plot leaving the border area. The plants were uprooted through auger and roots were washed with cleaned water. The root length was measured from the collar region upto the tip of tap root.

b) Root dry weight: Root dry weight was measured at 30 DAS. For this same plants as sampled for root length measurement were used. They were sun-dried and kept in an electric oven at 70°C for complete drying. The weight was expressed as g/plant.

c) Plant height: Plant height was measured at pod development stage at 30 DAS. Five plants were randomly selected from the central rows of the plot for height. Height was measured from the ground level to the tip of the top most leaf.

d) Number of flowers: Number of flowers was measured at full flowering stage at 75 DAS. Three plants were randomly selected from the central rows of the plot. The flowers were counted in each plant and average of three plant for each replication was noted.

e) Leaf chlorophyll content: Total leaf chlorophyll content in mature leaves was measured at first flower stage (30 DAS) by homogenizing 0.5 g fresh tissue in solution of 15 ml of 80% acetone and 20% water. The solution was filtered and absorbance was recorded at 645 and 663 nm through UV visible spectrophotometer (UV-2100). The absorbance values were substituted to calculate total chlorophyll (Chl. a+b) content expressed in mg g^{-1} on fresh weight basis.

$$\text{Total chlorophyll (mg g}^{-1}\text{)} = (8.05 \times A_{663} + 20.29 \times A_{645}) \times (V/W) \times (1/1000)$$

f) Plant population: Plant population/plot was recorded at 30 DAS and at maturity by counting the plants at centers of each plot using 1 m^2 (1 m x 1 m) quadrat and then multiplying with 25 to calculate the number of plants/plot.

g) Number of primary branches and leaves: Five plants that were randomly selected and tagged from the central rows for measuring height, were also used for measuring number of primary branches and leaves at 30 DAS.

h) Plant fresh weight and dry matter accumulation: For dry matter sampling, three plants were randomly selected from each plot in each observation at periodic intervals of 55 and 90 DAS from side rows (leaving border rows) excluding the net plot area. The sampled plants were dried in hot air oven at 70°C for 72 hours. Dry weight was expressed in g/plant.

i) Leaf length: Leaf length was measured at pod development stage at 30 DAS. Five plants sampled for root and nodule parameters were selected for leaf length measurements. Five leaves were selected and the length were measured from tip of leaf to the collar region of leaf.

j) Number and dry weight of nodules: The sampled plants which were used for measuring above and below ground dry matter accumulation were also used for measuring number of nodules/plant. Accordingly the numbers of nodules were collected and oven dried at 70°C for 72 hours and dry weight of nodules/plant was obtained.

k) Observation on days taken to initiation of flowering and maturity: The observations were made in every day and date of the initiation of first flush of flowering and maturity were recorded across all the treatments to know the effect of biofertilizers on flowering and maturity.



Fig. 2.4. Crop at podding stage

2.5. Yield attributes and yield

a) 100 seed weight: Grains were drawn randomly from the pods harvested from net plot and 1000 seeds were counted, weighed and expressed as 100 grain weight in g.

b) Grain yield: Grain yield was recorded from the net plot area (4 m x 4 m) from each plot. Then, seeds were dried and cleaned produce after threshing and weight was taken. Samples of crop were harvested and threshed manually from the central net areas for yield assessment. Finally, they were converted to q/ha. Yield was expressed at 12% moisture content.

c) Biomass (stover + husk) yield: The mungbean stalks of the net plot area were cut near the ground level with the help of sickle and weighed after sun drying. The weight of the husk was recorded from the pods and also added to the stover yield.



Fig. 2.5. Crop at maturity stage

2.6. Gravimetric moisture content

Gravimetric moisture content was determined by collecting field moist soils with soil cores at 0-20 cm soil depth. The collected soil sample was immediately kept in an aluminium moisture box (weighed moisture box) and wrapped with a cotton cloth to protect it from evaporation. The moisture boxes were transferred to the laboratory after sampling and fresh weight was recorded. Subsequently, aluminium box filled with soil was oven-dried at 105°C for 72 h. The weight of aluminium boxes was deducted from the fresh and dry weight of sample in the respective treatment. Finally, gravimetric moisture content and volumetric moisture content were estimated using following formulae:

Gravimetric moisture content (%) =

$$\frac{\text{Wet weight of soil sample (g)} - \text{Dry weight of soil sample (g)}}{\text{Dry weight of soil sample (g)}} \times 100$$

2.7. Soil Properties at Harvest

The soil organic carbon was SOC was estimated using 1 g soil in 20 ml sulphuric acid, 10 ml potassium dichromate as an oxidizer, and ferrous ammonium sulphate as titrating agent. Available N was estimated by the alkaline permanganate method using the Kjeldahl distillation set. Briefly, 20 g of soil was added to 20 ml of water, 100 ml of 0.32% potassium permanganate, and 100 ml of 2.5% sodium hydroxide in the distillation flask. Thereafter, 20 mL of 2% boric acid and 2-3 drops of methyl red indicator were kept in a beaker. Then soil solution mixture was distilled in a distillation set and ammonia gas was collected into boric acid solution. The distillate was titrated against 0.02 N sulphuric acid. Olsen's extractant (pH 8.5) was utilized for the estimation of available P using spectrophotometric blue color intensity at 660 nm wavelength. A 2.5 g soil was dipped into 50 ml of 0.5 M sodium bicarbonate and a pinch of activated charcoal and shaken for 30 min. Then the solution was filtered with Whatman No. 1 filter paper and 5 mL clear filtrate was transferred to a 25 ml volumetric flask. Finally, 5 ml of 1.5% ammonium molybdate solution was added to it and made the volume to 25 ml to read color intensity. Available K was determined using a 5 g soil sample and 25 ml of 1 N ammonium acetate (pH 7.0) was added to it. The soil pH and electrical conductivity (soil-to-water proportion of 1:2.5) was assessed utilizing standard method.

2.8. Grain analysis for macro nutrients

The phosphorus and potassium content in grains were estimated at harvest. The methods for estimation of nutrient contents were spectrophotometric estimation of vanodomolybdate yellow color for phosphorus, flame photometric estimation for potassium. Flame photometric method was utilized to estimate the calcium and magnesium content in grain. A 0.5 g mungbean grain was digested with di-acid digestion followed by made up volume upto 50 ml. A standard calcium solution on 25 ppm, 50 ppm, 75 ppm and 100 ppm using calcium carbonate. Similarly, a standard calcium solution on 25 ppm, 50 ppm, 75 ppm and 100 ppm using magnesium carbonate.

2.9. Economic analysis: The economic analysis in terms of gross and net returns and benefit: cost ratio (net returns per rupee invested) was worked out on the basis of existing rate of inputs and output. Total cost included in the cost of input such as seeds, fertilizers, irrigation and various cultural operations such as ploughing, sowing, weeding, harvesting, threshing etc. The rental value of land was also considered in the cost of cultivation. Minimum support of price of mungbean (year 2023 : INR 8558.00/q) was considered for calculating the gross return. Returns were calculated by using the following formula expression

1. Gross returns = Value of the seed + Value of stover
2. Net returns = Gross returns – Total costs of cultivation
3. Benefit: cost ratio = Gross returns/Total cost of cultivations

2.10. Statistical analysis

The data recorded for different parameters were analysed under randomized block design. The analysis of the data was tested by ANOVA using the statistical software package OPSTAT. The least square means (LS Means) of respective treatments obtained in analysis were compared by Least Significant Difference (LSD) at $P < 0.05$.

The results of the field experiment entitled “**Comparative analysis of fertilizers of LCB and other existing products in mungbean (Product: ‘Navyakosh’)**” has been summarized in this chapter with the help of appropriate tables and diagrams. The observations pertaining to crop growth, yield, nutrient status and economics were statistically analysed and the significance of results were verified.

3.1. Above-ground plant parameter

Applied nutrients management practices in mungbean resulted non-significant impacts on plant population. Further, significant results about plant height, branches/plant, leaves/plant, leaf length, plant fresh weight and plant dry weight recorded from Navyakosh applied treatment. This treatment recorded superior results about plant height (24.6%), branches/plant (11.5%), leaves/plant (14.6%), leaf length (18.2 %), plant fresh weight (21.4%) and plant dry weight (13.1%) over control treatment.

Table 3.1.: Influence of different nutrients management practices on above ground observations in mungbean.

Treatment	Plant Population	Plant height (cm)	Branches /plant	Leaves/ plant	Leaf length (cm)	Plant fresh wt. (g)	Plant dry wt. (g)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	30.8	57.7	2.37	27.0	6.83	191.6	15.4
Navyakosh 375 kg/ha at the time of land preparation	31.3	58.2	2.42	27.5	6.96	196.0	15.5
Farmyard manure 5 t/ha	28.8	51.8	2.25	25.0	6.24	172.1	14.4
Control	27.8	46.7	2.17	24.0	5.89	161.5	13.7
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	29.3	55.7	2.29	26.3	6.60	185.8	15.2
C.D.	N/A	5.83	0.16	2.19	0.53	16.4	1.26
S.E.(m)	0.9	1.87	0.05	0.70	0.17	5.27	0.41
C.V.	5.9	6.93	4.40	5.40	5.23	5.81	5.47

3.2. Below-ground plant parameter

The root length, nodules number/plant and dry weight of nodules/plant were not significantly affected by the nutrient management practices that were investigated. Meanwhile, executed Navyakosh had beneficial impacts on both the fresh and dried weight of

the roots. Navyakosh tested treatment had 34% greater root fresh weight and 13.5% greater root dry weight compared to control treatment.

Table 3.2.: Influence of different nutrients management practices on below ground observations of mungbean

Treatment	Root length (cm)	Root fresh wt. (g)	Root Dry wt. (g)	No. of nodules /plant	Nodules Dry wt. (g)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	16.3	9.64	1.23	29.6	0.090
Navyakosh 375 kg/ha at the time of land preparation	16.3	9.73	1.26	26.1	0.078
Farmyard manure 5 t/ha	16.0	8.56	1.17	24.3	0.063
Control	16.9	7.26	1.11	29.2	0.075
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	16.4	9.17	1.20	33.6	0.065
C.D.	N/A	0.95	0.10	N/A	N/A
S.E.(m)	0.71	0.30	0.03	2.1	0.007
C.V.	8.65	6.86	5.10	15.0	18.7

3.3.: Soil moisture, chlorophyll content, flowers and days to maturity

None of the investigated nutrient management practices had significant impacts on the moisture content of the soil. While, chlorophyll content and number of flowers/plant significantly influenced by the investigated practices. In Navyakosh applied treatment plant has 14.9% more chlorophyll content compared to control treatment. Subsequently, Navyakosh applied treatment also produces 21.5% more number of flowers in comparison of control treatment. However, applied nutrients management practices also significantly affecting the crop maturation time taken by crop. The maturity of Navyakosh applied treatment recorded 2 days earlier compared to control treatment.

Table 3.3.: Influence of different nutrients management practices on soil moisture, chlorophyll content, flower number and days to maturity of mungbean

Treatment	Soil moisture (%)	Chlorophyll (mg/g)	Flowers No./plant	Days to maturity
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	19.1	39.1	26.6	65.0
Navyakosh 375 kg/ha at the time of land preparation	20.8	40.7	27.1	60.8

Farmyard manure 5 t/ha	19.8	36.3	23.2	62.3
Control	19.0	35.4	22.3	62.8
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	18.3	38.1	26.1	65.5
C.D.	N/A	3.57	3.5	1.1
S.E.(m)	1.5	1.15	1.1	0.3
C.V.	15.7	6.04	9.0	1.1



Fig. 3.1. Crop with full podding stage

3.4.: Yield attributes and yield

Yield attributing characters and yield of mungbean varied according to applied different interventions of nutrients management. Further, pod/plant, pod weight/plant and pod length recorded highest from second treatment, which resulted 21.6% more pod/plant, 21.1% more pod weight/plant and 21.3% greater pod length over control treatment. Subsequently, number of seed/pod, seed weight/pod and 100 seed weight also recorded maximum from Navyakosh applied treatment. It resulted 24.5% higher number of seed pod⁻¹, 23.3% greater seed weight pod⁻¹ and 24.8% greater weight of 100 seed in comparison of control treatment. Finally, the highest yield also recorded from Navyakosh tested treatment, which has 29.0% greater grain yield over control treatment and 5.0% greater yield over recommended chemical fertilizers applied treatment.

Table 3.4.: Influence of different nutrients management practices on yield attributing characters and yield of mungbean

Treatment	Pod/plant	Pod weight/plant	Pod length (cm)	No. of seed/pod	Seed wt./pod (g)	100 Seed wt. (g)	Yield (q/ha)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	20.0	13.0	7.39	7.98	0.36	3.95	7.16
Navyakosh 375 kg/ha at the time of land preparation	20.3	13.2	7.53	8.13	0.37	4.03	7.51
Farmyard manure 5 t/ha	17.4	11.3	6.44	6.95	0.31	3.44	6.46
Control	16.7	10.9	6.21	6.53	0.30	3.23	5.82
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	19.6	12.7	7.25	7.65	0.35	3.79	7.00
C.D.	2.63	1.45	0.88	1.15	0.05	0.57	0.57
S.E.(m)	0.84	0.47	0.28	0.37	0.02	0.18	0.18
C.V.	8.99	7.62	8.13	9.89	10.19	9.87	5.38

3.5: Soil chemical properties

Soil of the experimental field were found to be alkaline in nature with values of pH ranging from 8.18 to 8.27 and of electrical conductivity values varied under range of 1.36 to 1.55 ds m⁻¹. Subsequently, the applied practices of nutrients managements recorded non-significant impacts on pH, electrical conductivity, organic carbon and nitrogen content of investigated soil. Furthermore, investigated practices of nutrients management recorded significant impacts on phosphorus and potassium content of soil. The phosphorus content of soil recorded 30.9% greater and potassium content of soil recorded 28.7% greater in Navyakosh applied treatment over recommended dose of fertilizers applied treatment.

Table 3.5.: Influence of different nutrients management practices on chemical properties of soil

Treatment	pH	EC (dS/m)	OC (%)	N (kg/ha)	P (kg/ha)	K (kg/ha)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	8.18	1.55	0.31	94.1	11.0	178.9
Navyakosh 375 kg/ha at the time of land preparation	8.27	1.46	0.31	103.5	14.4	230.3
Farmyard manure 5 t/ha	8.27	1.54	0.28	97.2	12.5	211.5
Control	8.27	1.43	0.24	97.2	10.3	165.1
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	8.20	1.36	0.28	109.8	11.2	184.6
C.D.	N/A	N/A	N/A	N/A	1.66	31.8
S.E.(m)	0.08	0.08	0.02	7.77	0.53	10.2
C.V.	1.99	10.9	10.3	15.5	8.96	10.5

3.6: Nutrients content (%) in plant

Different nutrients management practices recorded statistically non-significant impacts on nitrogen and potassium content (%) of plant. While, nutrients management practices are significantly influencing the phosphorus content (%) of plant. Further, the Navyakosh applied treatment recorded 13.8% more p content in plant in comparison of control treatment.

Table 3.6.: Influence of different nutrients management practices on N, P and K content of mungbean plant

Treatment	N (%)	P (%)	K (%)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	2.15	0.26	1.13
Navyakosh 375 kg/ha at the time of land preparation	2.31	0.28	1.15
Farmyard manure 5 t/ha	2.11	0.23	1.08
Control	2.09	0.20	1.01
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	2.13	0.26	1.12
C.D.	N/A	0.02	N/A
S.E.(m)	0.103	0.01	0.04
C.V.	9.54	6.40	6.51

3.7: Nutrients content (%) in seed

Implementation of diverse nutrients management practices recorded non-significant impacts on N and K content (%) of seed. While, investigated practices of nutrients management recorded significant impacts on P (%) content of seed. P content of seed recorded 16.3% more p content in seed over control treatment.

Table 3.7.: Influence of different nutrients management practices on N, P and K content of mungbean seed

Treatment	N (%)	P (%)	K (%)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	3.66	0.48	0.70
Navyakosh 375 kg/ha at the time of land preparation	3.73	0.50	0.73
Farmyard manure 5 t/ha	3.51	0.46	0.66
Control	3.42	0.43	0.65
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	3.59	0.47	0.67
C.D.	N/A	0.04	N/A
S.E.(m)	0.13	0.01	0.02
C.V.	7.33	6.04	6.00

3.8: Economics

In current investigation the monetary net return (INR/ha) and B:C ratio (%) of applied practices were significantly influenced by different nutrients management practices. The monetary net return (INR/ha) recorded 12.6% greater in Navyakosh applied treatment over recommended dose of fertilizers applied treatment. Similarly, B:C ratio (%) also recorded highest from Navyakosh applied treatment, which has 18.1% greater results over recommended dose of fertilizers applied treatment.

Table 3.8.: Influence of different nutrients management practices on economics of different treatments

Treatment	Net return (INR/ha)	B:C ratio (%)
Recommended chemical fertilizer (NPK 20:60:40 kg/ha)	34,278	1.27
Navyakosh 375 kg/ha at the time of land preparation	38,586	1.50
Farmyard manure 5 t/ha	9,112	0.20
Control	28,613	1.35
Integrated nutrient management (50% NPK + Farmyard manure 5 t/ha)	10,787	0.22
C.D.	4,891	0.17
S.E.(m)	1,570	0.06
C.V.	13	12.0



Fig. 3.2. Visit of Scientist and LCB representative at ICAR-IIPR, Kanpur

3.9. Most effective rate

The most effective nutrients management practice based on the statistically analysed data on growth, yield, economics, soil properties and nutrients concentration of plant and seed recorded 'Navyakosh, which is applied during land preparation at 375 kg/ha and also recorded superior over control and other nutrients management practices including recommended dose of fertilizers.



Fig. 3.3. Navyakosh 375 kg/ha

Impacts of different nutrients management practices on growth and yield of mungbean has been clearly brought out in this investigation. Perusal of the data revealed that different nutrients management practices recorded significant impacts on growth observations, yield attributing characters and yield of plant. Applied biofertilizer recorded positive interactions with plant and roots, which resulted improvements in growth and yield of crop (Jeffries et al. 2003). Biofertilizer enhanced the growth and development of mungbean in several ways such as by fixing atmospheric nitrogen, solubilizing phosphorus and potassium, producing phytohormones, and suppressing different stresses and plant pathogens (Maçik et al., 2020; Mushtaq et al., 2021). Furthermore, the application of biofertilizers also revealed that the crop matured more quickly because of its high nutrient use efficiency and ability to withstand under stressful conditions. This allows the plants to complete their life cycle quickly, without experiencing any hindrance (Shaheen et al., 2013).

Conventional fertilizer application has several drawbacks such as environmental pollution, unbalanced soil fertility, emergence of new pests, reduction of soil fertility and productivity. The use of biofertilizers has been shown to be a viable substitute in this situation for sustaining agricultural productivity in comparison of conventional fertilizers (Chakraborty and Akhtar, 2021).

Among the different nutrients management practices ‘Navyakosh’ (applied during land preparation at 375 kg/ha) proved better in terms of growth, yield, soil properties and economics (net returns and B:C ratio).

The salient findings of the experiment are summarized as follows:

- ✱ ‘Navyakosh’ at 375 kg/ha resulted in higher grain yield by 29% and 5% than control and recommended chemical fertilizers applied practices, respectively.
- ✱ ‘Navyakosh’ at 375 kg/ha also recorded superior below and above ground growth parameters over control and recommended chemical fertilizers.
- ✱ Nutritional properties of soil, plant, and seed positively responded with Navyakosh application over control and rest of practices.
- ✱ The net returns were higher with ‘Navyakosh’ at 375 kg/ha over control and recommended chemical fertilizers by 34.9% and 12.6%, respectively.
- ✱ B:C ratio of Navyakosh applied practices also recorded greater by 11.1% and 18.1% than control and recommended chemical fertilizers applied practices, respectively.

Inference

Study indicated that ‘Navyakosh’ @ 375 kg/ha resulted in higher yield and profitability in mungbean, and can be recommended in mungbean.

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